

DATA SET
DS1403

23. Telecommuting As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

Salary	Workdays at Home per Week			
	Less Than One	At Least One, but Not All	All at Home	
Under \$50,000	38	16	14	68
\$50,000–\$74,999	54	26	12	92
\$75,000–\$99,999	35	22	9	66
Above \$100,000	33	29	12	74
sum	160	93	47	300

- a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.
- b. Use Table 5 in Appendix I to approximate the p -value for this test of hypothesis. Does the p -value confirm your conclusions from part a?

E_{ij}			
< 50,000	$\frac{68 \times 160}{300} = 36.2$	$\frac{68 \times 93}{300} = 21.08$	$\frac{68 \times 47}{300} = 10.65$
50,000	$\frac{92 \times 160}{300} = 49$	$\frac{92 \times 93}{300} = 28.52$	$\frac{92 \times 47}{300} = 14.41$
75,000	$\frac{66 \times 160}{300} = 35.2$	$\frac{66 \times 93}{300} = 20.46$	$\frac{66 \times 47}{300} = 10.34$
100,000	$\frac{74 \times 160}{300} = 39.47$	$\frac{74 \times 93}{300} = 22.94$	$\frac{74 \times 47}{300} = 11.59$

a. H_0 : factors are independent.
 H_1 : factors are dependent.

$$\alpha = 0.05, \quad df = (3-1)(4-1) = 6.$$

$$\chi^2_{STAT} = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{0.05, 6} = 12.592$$

$$\chi^2_x = 6.449, \quad RR: \{ \chi^2 \mid \chi^2 > 12.592 \}$$

$6.449 < 12.592 \Rightarrow$ non-reject H_0 .

There isn't sufficient evidence to say that salary is dependent on number of workday at home.

b. $\chi^2_{0.1} = 10.645 > 6.449$
 $\Rightarrow 0.05 < 0.1 < p\text{-value}$
 $\Rightarrow$ non-reject H_0 .
 same result